Measurement of Laser Beam Waist Using the Knife-Edge Method

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Knife-edge method

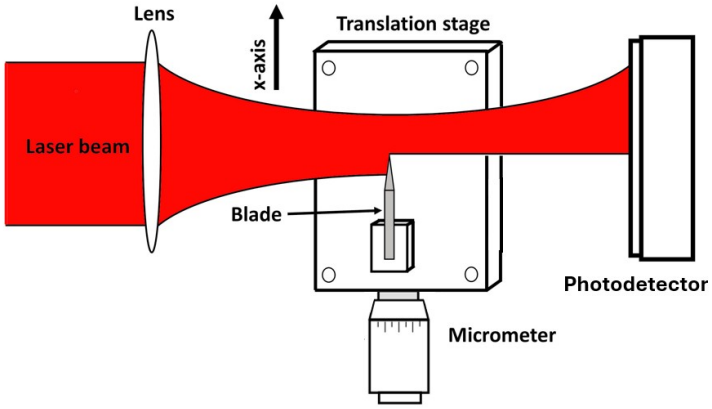

Figure 1. A scheme for the measurement of laser beam waist using the knife-edge technique.

A common laboratory technique for measuring the beam waist of a Gaussian beam is the knife-edge method. The intensity distribution in the x–y plane is

$$I(x,y) = I_0 \exp\left[-\frac{(x-x_0)^2+(y-y_0)^2}{w_0^2}\right], \tag{1}$$

where w_0 is the beam waist, I_0 is the peak intensity, and (x_0, y_0) is the beam center.

In this method, a blade is scanned along the x-axis, partially blocking the beam. The transmitted power is obtained by integrating the intensity over the unblocked region:

$$P(x) = \int_x^\infty \int_{-\infty}^\infty I(x',y')\,dy'\,dx'. \tag{2}$$

Assuming a Gaussian intensity profile,

$$I(x',y') = I_0 \exp\left(-\frac{(x'-x_0)^2+(y'-y_0)^2}{w_0^2}\right), \tag{3}$$

the integration over y' gives

$$\int_{-\infty}^\infty \exp\left(-\frac{(y'-y_0)^2}{w_0^2}\right) dy' = w_0\sqrt{\pi}. \tag{4}$$

Thus,

$$P(x) = I_0 w_0 \sqrt{\pi} \int_x^\infty \exp\left(-\frac{(x'-x_0)^2}{w_0^2}\right) dx'. \tag{5}$$

With the substitution

$$u = \frac{x'-x_0}{w_0}, \quad dx' = w_0\,du,$$

this becomes

$$P(x) = I_0 w_0^2 \sqrt{\pi} \int_{\frac{x-x_0}{w_0}}^\infty e^{-u^2}\,du. \tag{6}$$

Using the identity

$$\int_a^\infty e^{-u^2}\,du = \frac{\sqrt{\pi}}{2}\left[1-\operatorname{erf}(a)\right], \tag{7}$$

we obtain

$$P(x) = \frac{I_0 \pi w_0^2}{2} \left[1 - \mathrm{erf}\left(\frac{x - x_0}{w_0} \right) \right]. \tag{8}$$

More generally, allowing for an offset and maximum power scaling, this can be written as [1, 2]

$$P(x) = P_0 + \frac{P_{\max}}{2} \left[1 - \mathrm{erf}\left(\frac{x - x_0}{w_0} \right) \right]. \tag{9}$$

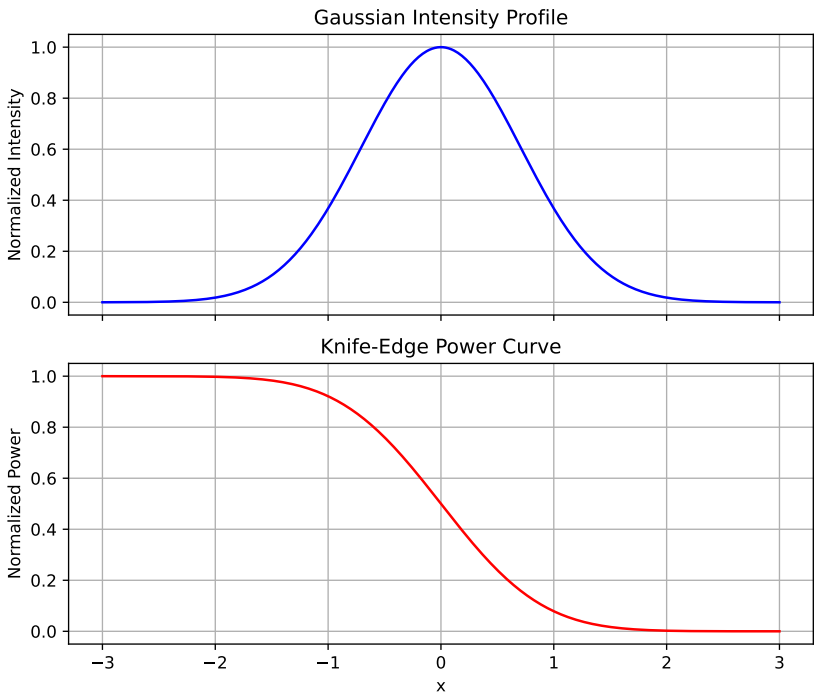

Figure 2. Normalized Gaussian beam intensity and knife-edge power as functions of the x-position. The top panel shows the normalized intensity $I(x)/I_{\max}$, following a Gaussian profile centered at x_0. The bottom panel shows the normalized transmitted power $P(x)$ obtained using the knife-edge method, which follows an error-function dependence. Both curves share the same x-axis for direct comparison of the beam profile and the integrated power.

Python script

```python
"""
Estimate beam radius using knife-edge method
"""
import pandas as pd
import numpy as np
import math
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
from scipy import special
from sklearn.metrics import r2_score

data = pd.read_csv('vertical_beam.csv')

x = data['Position (mm)'].values
y = data['Power (mW)'].values

def model(x, x0, p0, p_max, w):
    return p0 + (p_max/2) * (1 - special.erf(np.sqrt(2)*(x0 - x)/w))

initial_guess = [7, 0, 0.34, 0.5]

params, _ = curve_fit(model, x, y, initial_guess)

yp = model(x, *params)

print(f"x0 = {params[0]:.2f} mm")
print(f"Background = {params[1]:.3f} mW")
print(f"Max power = {params[2]:.3f} mW")
print(f"Beam waist = {params[3]:.2f} mm")
print(f"R^2 = {r2_score(y, yp):.5f}")

plt.scatter(x, y, alpha=0.5)
plt.plot(x, yp)
plt.xlabel('Position (mm)')
plt.ylabel('Power (mW)')
plt.title('Knife-edge measurement')
plt.show()
```

References

[1] J. Arnaud, W. Hubbard, G. Mandeville, B. De la Claviere, E. Franke, and J. Franke, "Technique for fast measurement of gaussian laser beam parameters," *Applied optics*, vol. 10, no. 12, pp. 2775–2776, 1971.

[2] M. González-Cardel, P. Arguijo, and R. Díaz-Uribe, "Gaussian beam radius measurement with a knife-edge: a polynomial approximation to the inverse error function," *Applied optics*, vol. 52, no. 16, pp. 3849–3855, 2013.